Lecture Notes on

The Search for Extraterrestrial Intelligence

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These notes summarize the guest talk and live question-and-answer session given by Dr. Seth Shostak, Senior Astronomer at the SETI Institute, at our second meeting of the year. They follow his talk itself: why scientists expect that life is out there, why we cannot simply travel to it, how SETI actually looks, when he thinks we might succeed, and what he believes the aliens will turn out to be.

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1 What SETI Is, and What It Is Not

- **SETI** stands for the **Search for Extraterrestrial Intelligence**. Dr. Shostak is a Senior Astronomer at the SETI Institute, which is in Mountain View, not far from us. The job is to try to find intelligent aliens.
- **We have not found any.** Despite what shows up on television, there has been no discovery and there is no government cover-up. About a third of the public believes the government is hiding something, usually pointing to Roswell, New Mexico.
- **The Roswell story does not hold up.** A craft that crossed many light years would not fail in its final few hundred feet and crash into the desert. It does not make sense.
- **A real discovery could not stay hidden.** If aliens had been found it would be the biggest story in the world, and every science museum would display the proof. That proof is not there. This is Nobel Prize material, not something anyone keeps in a desk drawer.
- **Television has the opposite incentive.** Saying "we have not found anything" gets no viewers, so programs prefer "we found something and it is being kept quiet." That sells, but it is not true.
- **The search is international but low key.** The United States does the most; historically the Soviet Union searched, and there is some work in Italy, France, and Britain. Very few people on Earth earn a living looking for aliens, fewer than would fill a couple of rows of a classroom, because there is no obvious profit in it compared to, say, building faster computers.

2 Why We Think Life Is Out There

We believe it not because we have found it, but from two lines of reasoning.

2.1 There is a lot of real estate

- When Dr. Shostak was young, we did not know whether other stars even had planets. Now we do: roughly 80% of stars like the Sun have planets.
- Most planets are useless for life, with no oceans, no atmosphere, or no solid ground to stand on. Our own solar system is a good example: Jupiter and Saturn are not places you could visit or stand on.
- Even so, some fraction is promising. An estimated 20% to 30% of planets around other stars are somewhat Earth-like: roughly Earth-sized, with an atmosphere and the raw material needed for life.

2.2 From planets to life to intelligence

- Even if you are handed a million Earth-like planets, we do not know how many ever produce life. A great many things had to go right for it to begin on Earth.
- Where life does arise, most of it stays simple. Most life on Earth is microbial, things like bacteria. Only a very small fraction is intelligent.

- Life also took a long time to appear here: the first life arose about 3.5 billion years ago. And life is not aiming at intelligence, it is aiming at survival. Bacteria are superb survivors. So the universe may hold plenty of life that never becomes smart.

2.3 We are probably not special

- For intelligent aliens to be absent, Earth, and you, would have to be extraordinarily special.
- The history of science argues against that. Every time we assumed we were special, we were wrong. We once thought Earth was the only world; then we found the other planets of the solar system; now we know there are trillions of planets in our galaxy alone.
- Following that pattern, intelligent life is probably out there as well, and it probably thinks it is special too.

3 Why We Cannot Simply Go There

- The distances are enormous. The nearest other star system is about 4.5 light years away, and one light year is about 6×10^{12} miles.
- Our fastest rockets travel about 7 miles per second. That is fast for a trip across the Bay Area, but at that speed the nearest star is roughly 75,000 years away, one way.
- Rocketing off to the stars works in movies, not in reality. This is a limit of physics, not just of current engineering: it is very hard to build a rocket that reaches another star system within a human lifetime.
- For the same reason, aliens have probably not come here either. About 30% of Americans believe aliens are visiting, but there is no good evidence, and no museum has anything genuine to show. If it were true, there would be something to see.

4 How SETI Actually Searches

Because we cannot travel there and they have probably not visited, we look for evidence at a distance.

4.1 Signs of what they built

- One option is to look for very large structures an advanced civilization might build, for example enormous solar collectors in space, bigger than a planet. We have not seen any, but it is worth looking.

4.2 Listening for signals

- The main method is to listen for transmissions: radio, radar, or flashing lasers. These are all ways of sending bits of information from one place to another.
- For about 60 years, SETI has pointed large antennas at nearby star systems and tuned across millions of channels, listening for a signal. Nothing has been confirmed so far.

- **Telling artificial from natural:** we look for **narrow-band** signals, energy concentrated at a single spot on the dial, like a radio station. Natural sources such as quasars and pulsars are broadband and show up all across the dial. A narrow-band signal means a transmitter, which is the first step even before we understand any message.
- The real bottleneck is money, not technology. We already have the antennas and the receivers; there is simply very little funding, because there is no profit in it.

5 When Might We Succeed?

- Combine the number of nearby star systems worth checking with the rate at which our technology improves (Moore's law, which was coined here in Silicon Valley), and you can make an estimate.
- Dr. Shostak's estimate: we will find evidence of intelligent life within about 20 years. His standing bet is a cup of coffee. Either we find them within 20 years, or he buys you a coffee, so you cannot lose.
- The larger point: you are the first generation that even could find them. Julius Caesar had no chance, because he lacked both the knowledge and the technology. You have both.

6 What Will the Aliens Be Like?

- Probably not the television version with green skin and extra limbs. They might resemble us, since the human body plan, a head on top and limbs for moving and handling things, is a reasonable outcome of evolution, and other intelligent animals on Earth share features with us. But Dr. Shostak doubts it.
- **His bet: they will be machines.** The most important development likely to happen within your lifetime is that we will invent our successors, machines that can truly think. We do not have them yet, but we will, whether that is 10 years or 110 years away; on cosmic timescales those are the same short interval.
- Homo sapiens appeared only about a million years ago, and soon after, on that scale, we are inventing machines smarter than ourselves, which will eventually take the lead.
- **Smarter does not mean hostile.** Being more intelligent than the squirrels in your backyard does not make you want to wipe them out; there is no motive. A thinking machine need not want to destroy us.

7 Questions from the Audience

7.1 Do we filter target planets by physics, not just water?

- We generally assume a life-bearing planet needs water and an atmosphere, as Earth does. Water is especially good for chemistry: oceans host an enormous amount of random chemical mixing, and given enough time, millions of years, that can eventually produce a molecule that copies itself.

- Once you have self-replication you have life, because a copy makes more copies, and Darwinian evolution follows: the best copies survive, and complex organisms eventually appear. So habitable alien worlds are probably somewhat Earth-like.

7.2 How do you know a signal is from aliens and not from nature?

- By its bandwidth. We look for narrow-band signals confined to one spot on the dial, the hallmark of a transmitter. Natural cosmic sources are broadband and appear all across the dial.

7.3 How far away are the nearest aliens?

- A rough guess: if about 1 in a million Sun-like stars hosts an Earth-like planet with intelligent life, the nearest neighbors might be around 100 light years away.
- That is still enormous. One light year is about 6×10^{12} miles, and at our fastest rocket speeds even 4 light years would take about 75,000 years.

7.4 Do we send robots to find alien planets?

- No. We find exoplanets mostly from the ground, with help from space telescopes, by watching the star rather than the planet. An orbiting planet tugs on its star and makes it wobble slightly, and that wobble is detectable across great distances.
- The aliens themselves might turn out to be machines, but that is a separate point; we do not send robots out to hunt for planets.

7.5 How do we tell if an exoplanet has water?

- When a planet passes in front of its star, it dims the star slightly, about 1 part in 1000, like a tiny eclipse.
- During that transit, some starlight passes through the planet's atmosphere. Splitting that light into its colors with spectroscopy, much like using a prism, reveals what the atmosphere contains, including water.

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